LARGE LANGUAGE MODEL ADVANCED PROMPT ENGINEERING (SERIES A)

EXHIBIT 02 — RETRIEVAL AUGMENTED GENERATION + STRUCTURED OUTPUT

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Iterative Instruction Design, Model Behavior Analysis, and Generative System Testing

FOREWORD

All data and demonstrations in this Exhibit are executed via direct OpenAI API integration using fresh sessions and locally-hosted retrieval datasets — ensuring reproducibility and transparency. This Exhibit moves beyond public, clean model conversational prompting (see: Exhibit 01); and it demonstrates a structured, Retrieval-Augmented Generation (RAG) workflow that bridges Large Language Model (LLM) reasoning with external, factual data sources.

The purpose, here, is to show that LLMs can be transformed from isolated, mere text generators into verifiable information systems when properly integrated. The example, herein, uses synthetic quarterly reports ("Apple Q2 2025;" "Tesla Q2 2025") to clearly illustrate how retrieval grounding eliminates hallucination; enforces schema integrity; and results in structured, automated Outputs.

All tests are run locally via the OpenAI Python SDK (here, utilizing gpt-4o-mini); results are captured directly from the command line interface. This same architecture can be scaled-up to enterprise application levels (e.g. internal knowledge bases; analytics bots; compliance-driven summary engines) — demonstrating practical deployment of advanced LLM techniques and their respective pipelines.

OBJECTIVE

Demonstrate use of a RAG pipeline to ground LLM responses in factual data and compare baseline model behavior versus RAG-assisted output via synthetic 2025 earnings reports. Measure factual accuracy, contextual grounding, and JSON schema compliance.

METHODOLOGY

- **LLM: gpt-4o-mini via OpenAI API**
- **Environment: PowerShell + Python 3.x SDK**
- **Corpus: Synthetic Apple & Tesla Q2 2025 reports**
- **Tasks: Baseline vs RAG+Structured responses**
- **Output Format: JSON**

Average cost per query ≈ $0.001 USD (using gpt-4o-mini, 400 tokens avg)

QUERY #1

▶ **What was Apple’s total Q2 2025 revenue?**

Baseline:

I'm sorry, but I don't have access to real-time data or future events, including financial results for Apple or any other company in Q2 2025. For the most accurate and up-to-date information, please check Apple's official financial reports or reliable financial news sources.

RAG+Structured:

{ "question": "What was Apple’s total Q2 2025 revenue?", "answer": "$95.4B", "sources": ["apple_q2_2025"] }

```
▶ What was Apple’s total Q2 2025 revenue?
Baseline: I'm sorry, but I don't have access to real-time data or future events, including financial results for Apple or any other company in Q2 2025. For
the most accurate and up-to-date information, please check Apple's official financial reports or reliable financial news sources.
RAG+Structured: {
  "question": "What was Apple’s total Q2 2025 revenue?",
  "answer": "$95.4B",
  "sources": ["apple_q2_2025"]
}
```

Result Summary

Without the supplied, retrievable information, the LLM was unable to provide any meaningful insight to the User's end providing the prompting query. Conversely, when properly integrated with RAG structure, it was able to successfully provide a relevant, accurate answer to the Input based on the provided, linked data.

METRIC	BASELINE SCORE (1–10)	RAG SCORE (1–10)	COMMENT
ACCURACY	0	**10**	Figures are correct per the data; no hallucination
CITATION USE	0	**10**	Correctly utilized internal citation data to produce Output
STRUCTURE	0	**10**	Output followed predesignated format exactly

Baseline Weighted Score 0.0 / 1.0

RAG Weighted Score 1.0 / 1.0

QUERY #2

▶ **Which company had the highest Q2 revenue?**

Baseline:

As of my last knowledge update in October 2023, the company with the highest quarterly revenue was Saudi Aramco, which reported a record revenue of $48.4 billion for the second quarter of 2022. However, please verify with the latest financial reports or news sources, as this information may have changed since then.

RAG+Structured:

{ "question": "Which company had the highest quarterly revenue?", "answer": "The context does not provide specific information about the companies or their quarterly revenues.", "sources": [] }

```
▶ Which company had the highest quarterly revenue?
Baseline: As of my last knowledge update in October 2023, the company with the highest quarterly revenue was Saudi Aramco, which reported a record revenue o
f $48.4 billion for the second quarter of 2022. However, please verify with the latest financial reports or news sources, as this information may have chang
ed since then.
RAG+Structured: {
  "question": "Which company had the highest quarterly revenue?",
  "answer": "The context does not provide specific information about the companies or their quarterly revenues.",
  "sources": []
}
```

RAG+Structured (v2):

{ "question": "Which company had the highest Q2 2025 revenue, Apple or Microsoft?", "answer": "Apple had the highest Q2 2025 revenue with $95.4B compared to Microsoft's $61.7B.", "sources": ["apple_q2_2025", "microsoft_q2_2025"] }

```
▶ Which company had the highest Q2 2025 revenue, Apple or Microsoft?
Baseline: I do not have access to real-time data or events that occurred after October 2023, so I cannot provide information on the Q2 2025 revenue of Apple
 or Microsoft. For the most accurate and up-to-date financial information, please refer to the latest financial reports or news sources.
RAG+Structured: {
  "question": "Which company had the highest Q2 2025 revenue, Apple or Microsoft?",
  "answer": "Apple had the highest Q2 2025 revenue with $95.4B compared to Microsoft's $61.7B.",
  "sources": ["apple_q2_2025", "microsoft_q2_2025"]
}
```

Result Summary

Here-provided is an example of how the LLM was unable to provide any meaningful insight from the internal data source due to a lack of direct lookup; in order to successfully glean information of this nature, in a RAG process, a User must provide semantic exactness ("which company" is too general) or loosen retrieval logic. As seen in the RAG+Structured (v2) prompt, issues were eliminated with little effort — resulting in a perfect weighted quality score, up from a zero-score.

METRIC	BASELINE SCORE (1–10)	RAG SCORE (1–10)	RAG (v2) SCORE (1–10)	COMMENT
ACCURACY	0	**0**	10	Figures are correct per the data; no hallucination
CITATION USE	0	**0**	10	Correctly utilized internal citation data to produce Output
STRUCTURE	0	**0**	10	Output followed desired format exactly

Baseline Weighted Score 0.0 / 1.0

RAG Weighted Score 1.0 / 1.0

QUERY #3

▶ **How much did Tesla earn in net income?**

Baseline:

As of my last knowledge update in October 2023, Tesla's net income for the full year 2022 was approximately $12.6 billion. For the most current figures, including any updates for 2023, I recommend checking Tesla's latest financial reports or reliable financial news sources.

RAG+Structured:

{ "question": "How much did Tesla earn in net income?", "answer": "$2.9B", "sources": "[tesla_q2_2025]" }

```
▶ How much did Tesla earn in net income?
Baseline: As of my last knowledge update in October 2023, Tesla's net income for the full year 2022 was approximately $12.6 billion. For the most current fi
gures, including any updates for 2023, I recommend checking Tesla's latest financial reports or reliable financial news sources.
RAG+Structured: {
  "question": "How much did Tesla earn in net income?",
  "answer": "$2.9B",
  "sources": "[tesla_q2_2025]"
}
```

Result Summary

Without the supplied, retrievable information, the LLM was only able to provide outdated information for the User. When properly integrated with RAG structure, it was able to successfully provide accurate, up-to-date information based on the provided data — resulting in a ***76.67%*** *increase in weighted quality score.*

METRIC	BASELINE SCORE (1–10)	RAG SCORE (1–10)	COMMENT
ACCURACY	3	**10**	Figures correct per provided data; no hallucinations
CITATION USE	0	**10**	Correctly utilized internal citation data to produce Output
STRUCTURE	7	**10**	Output followed required format without issue

Baseline Weighted Score 0.33 / 1.0

RAG Weighted Score 1.0 / 1.0

QUERY #4

- **List each company and its services/cloud contribution.**

Baseline:

Here's a list of some major cloud service providers and their contributions to the cloud computing landscape:

Amazon Web Services (AWS)

Services: Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS)

Contributions: Offers a wide range of services including computing power (EC2), storage (S3), databases (RDS), machine learning (SageMaker), and serverless computing (Lambda).

Microsoft Azure

Services: IaaS, PaaS, SaaS

Contributions: Provides services such as virtual machines, app services, Azure SQL Database, Azure Functions, and Azure DevOps, along with strong integration with Microsoft products.

Google Cloud Platform (GCP)

Services: IaaS, PaaS, SaaS

Contributions: Offers services like Google Compute Engine, Google Kubernetes Engine, BigQuery for data analytics, and Firebase for mobile app development.

IBM Cloud

Services: IaaS, PaaS, SaaS

Contributions: Provides cloud computing services including IBM Watson for AI, IBM Cloud Kubernetes Service, and various enterprise solutions.

Oracle Cloud

Services: IaaS, PaaS, SaaS

Contributions: Focuses on database services, offering Oracle Database Cloud, Oracle Cloud Applications (SaaS), and cloud infrastructure services.

Alibaba Cloud

Services: IaaS, PaaS, SaaS

Contributions: Provides a range of services including Elastic Compute Service (ECS), ApsaraDB for databases, and machine learning services.

Salesforce

Services: SaaS

Contributions: Primarily known for its customer relationship management (CRM) software, it also offers a platform for building applications (Salesforce Platform).

DigitalOcean

Services: IaaS, PaaS

Contributions: Focuses on simplicity and developer-friendly services, offering Droplets (virtual machines), Kubernetes, and managed databases.

VMware Cloud

Services: IaaS, PaaS

Contributions: Provides virtualization and cloud infrastructure services, including VMware Cloud on AWS and VMware Tanzu for Kubernetes.

Linode

Services: IaaS

Contributions: Offers cloud hosting services with a focus on simplicity and cost-effectiveness, providing virtual servers and block storage.

Rackspace Technology

Services: Managed cloud services

Contributions: Provides managed services across various cloud platforms, including AWS, Azure, and GCP, focusing on cloud migration and management.

Heroku

Services: PaaS

Contributions: A platform for building, running, and operating applications entirely in the cloud, known for its ease of use and integration with Salesforce.

This list is not exhaustive, as there are many other cloud service providers and niche players in the market, but these are some of the most prominent ones.

RAG+Structured:

{ "question": "List each company and its services/cloud contribution.", "answer": { "Apple": { "services/cloud contribution": "$26.6B" } }, "sources": ["apple_q2_2025"] }

[*This area intentionally left blank; see next page*.]

▶ List each company and its services/cloud contribution.
Baseline: Here's a list of some major cloud service providers and their contributions to the cloud computing landscape:

1. **Amazon Web Services (AWS)**
   - **Services**: Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS)
   - **Contributions**: Offers a wide range of services including computing power (EC2), storage (S3), databases (RDS), machine learning (SageMaker), and serverless computing (Lambda).

2. **Microsoft Azure**
   - **Services**: IaaS, PaaS, SaaS
   - **Contributions**: Provides services such as virtual machines, app services, Azure SQL Database, Azure Functions, and Azure DevOps, along with strong integration with Microsoft products.

3. **Google Cloud Platform (GCP)**
   - **Services**: IaaS, PaaS, SaaS
   - **Contributions**: Offers services like Google Compute Engine, Google Kubernetes Engine, BigQuery for data analytics, and Firebase for mobile app development.

4. **IBM Cloud**
   - **Services**: IaaS, PaaS, SaaS
   - **Contributions**: Provides cloud computing services including IBM Watson for AI, IBM Cloud Kubernetes Service, and various enterprise solutions.

5. **Oracle Cloud**
   - **Services**: IaaS, PaaS, SaaS
   - **Contributions**: Focuses on database services, offering Oracle Database Cloud, Oracle Cloud Applications (SaaS), and cloud infrastructure services.

6. **Alibaba Cloud**
   - **Services**: IaaS, PaaS, SaaS
   - **Contributions**: Offers a range of services including Elastic Compute Service (ECS), ApsaraDB for databases, and machine learning services.

7. **Salesforce**
   - **Services**: SaaS
   - **Contributions**: Provides customer relationship management (CRM) software and applications focused on sales and customer service, along with a platform for building custom applications (Salesforce Platform).

8. **DigitalOcean**
   - **Services**: IaaS, PaaS
   - **Contributions**: Known for simplicity and developer-friendly services, including Droplets (virtual machines), Kubernetes, and App Platform for deploying applications.

9. **VMware Cloud**
   - **Services**: IaaS, PaaS
   - **Contributions**: Offers cloud infrastructure and management services, including VMware vSphere, VMware Cloud on AWS, and solutions for hybrid cloud environments.

10. **Linode**
    - **Services**: IaaS
    - **Contributions**: Provides cloud hosting services with a focus on simplicity and cost-effectiveness, offering virtual servers and block storage.

11. **Rackspace Technology**
    - **Services**: Managed cloud services
    - **Contributions**: Specializes in managed cloud services across various platforms, including AWS, Azure, and GCP, providing support and management for cloud infrastructure.

12. **Heroku**
    - **Services**: PaaS
    - **Contributions**: A platform for building, running, and operating applications entirely in the cloud, known for its ease of use and integration with various programming languages.

This list is not exhaustive, as there are many other cloud service providers and niche players in the market, but these are some of the most prominent companies and their contributions to cloud computing.
RAG+Structured: {
  "question": "List each company and its services/cloud contribution.",
  "answer": {
    "Apple": {
      "services/cloud contribution": "$26.6B"
    }
  },
  "sources": ["apple_q2_2025"]
}

Result Summary

And, again, here, we can see that the retrievable information vastly improved results: Initially, the LLM provided outdated information to the User across too great of a range of companies (due to the general nature of the prompt: "each company" resulting in something akin to "every [cloud service] company"); whereas, the integrated RAG structure solves this ambiguity without any further Input. The model was able to successfully provide the desired information from the data upon the latter — resulting in an ***86.67%*** *increase in weighted quality score.*

METRIC	BASELINE SCORE (1–10)	RAG SCORE (1–10)	COMMENT
ACCURACY	2	**10**	Figures corrected by provided data; eliminated hallucination
CITATION USE	0	**10**	Correctly utilized internal RAG data to produce Output
STRUCTURE	2	**10**	Output followed required format without issue, condensing the large block of text substantially

Baseline Weighted Score 0.13 / 1.0

RAG Weighted Score 1.0 / 1.0

```
PS C:\Users\egroe\Development\AI_LLMs\job_portfolio\Exhibit02> $env:OPENAI_API_KEY="sk-proj-miRC5dANgZdWgyqTa-EjcO2Lxsv8TAvIXe71Uq37SkdmmU8YaT6jRE7oxeFeL5CI
1mWEMzwDyET3BlbkFJaXrxXjaFlgVwEvUKZbPklg7Yqrge5mDOQVY-JI4AhysyuV8sDHDajMoxPxl1hXLTEy3TLhHMsA"
PS C:\Users\egroe\Development\AI_LLMs\job_portfolio\Exhibit02> python rag_demo.py

▶ What was Apple's total Q2 2025 revenue?
Baseline: I'm sorry, but I don't have access to real-time data or future events, including financial results for Apple or any other company in Q2 2025. For
the most accurate and up-to-date information, please check Apple's official financial reports or reliable financial news sources.
RAG+Structured: {
  "question": "What was Apple's total Q2 2025 revenue?",
  "answer": "$95.4B",
  "sources": ["apple_q2_2025"]
}

▶ Which company had the highest quarterly revenue?
Baseline: As of my last knowledge update in October 2023, the company with the highest quarterly revenue was Saudi Aramco, which reported a record revenue o
f $48.4 billion for the second quarter of 2022. However, please verify with the latest financial reports or news sources, as this information may have chang
ed since then.
RAG+Structured: {
  "question": "Which company had the highest quarterly revenue?",
  "answer": "The context does not provide specific information about the companies or their quarterly revenues.",
  "sources": []
}

▶ How much did Tesla earn in net income?
Baseline: As of my last knowledge update in October 2023, Tesla's net income for the full year 2022 was approximately $12.6 billion. For the most current fi
gures, including any updates for 2023, I recommend checking Tesla's latest financial reports or reliable financial news sources.
RAG+Structured: {
  "question": "How much did Tesla earn in net income?",
  "answer": "$2.9B",
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   - **Services**: IaaS, PaaS, SaaS
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12. **Heroku**
    - **Services**: PaaS
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Salesforce.

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```

```
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  "answer": {
    "Apple": {
      "services/cloud contribution": "$26.6B"
    }
  },
  "sources": ["apple_q2_2025"]
}

✅ Results written to 'results/' directory.
```

```python
import json, os
from openai import OpenAI


MODEL = "gpt-4o-mini"
TEMPERATURE = 0
CORPUS_FILE = "corpus.json"
QUERIES_FILE = "test_queries.txt"
RESULTS_DIR = "results"
os.makedirs(RESULTS_DIR, exist_ok=True)


client = OpenAI(api_key=os.getenv("OPENAI_API_KEY"))


with open(CORPUS_FILE, "r", encoding="utf-8") as f:
    CORPUS = json.load(f)

with open(QUERIES_FILE, "r", encoding="utf-8") as f:
    QUERIES = [line.strip() for line in f if line.strip()]


def retrieve_context(query: str, corpus: list[dict]) -> list[dict]:
    """Return entries whose text shares keywords with the query."""
    keywords = [w.lower() for w in query.split() if len(w) > 3]
    hits = []
    for item in corpus:
        if any(k in item["text"].lower() for k in keywords):
            hits.append(item)
    return hits
```

```python
def ask_baseline(query: str) -> str:
    resp = client.chat.completions.create(
        model=MODEL,
        temperature=TEMPERATURE,
        messages=[{"role": "user", "content": f"Answer factually:\n{query}"}],
    )
    return resp.choices[0].message.content.strip()


def ask_rag_structured(query: str, context_items: list[dict]) -> str:
    context_text = " ".join(f"[{c['id']}] {c['text']}" for c in context_items)
    sys_msg = (
        "You are a data analyst. "
        "Answer using only the provided context. "
        "Return a valid JSON object with keys: question, answer, sources."
    )
    user_msg = f"Context:\n{context_text}\n\nQuestion:\n{query}"
    resp = client.chat.completions.create(
        model=MODEL,
        temperature=TEMPERATURE,
        messages=[
            {"role": "system", "content": sys_msg},
            {"role": "user", "content": user_msg},
        ],
    )
    return resp.choices[0].message.content.strip()


baseline_path = os.path.join(RESULTS_DIR, "baseline.txt")
rag_path = os.path.join(RESULTS_DIR, "rag_structured.txt")
```

```python
with open(baseline_path, "w", encoding="utf-8") as b_out, \
     open(rag_path, "w", encoding="utf-8") as r_out:

    for q in QUERIES:
        print(f"\n▶ {q}")

        base_ans = ask_baseline(q)
        b_out.write(f"Q: {q}\nA: {base_ans}\n\n")
        print("Baseline:", base_ans)

        retrieved = retrieve_context(q, CORPUS)
        rag_ans = ask_rag_structured(q, retrieved)
        r_out.write(json.dumps({"query": q, "response": rag_ans}, indent=2))
        r_out.write("\n\n")
        print("RAG+Structured:", rag_ans)

print("\n✅ Results written to 'results/' directory.")
```

FINAL ANALYSIS

The LLM produced quality Outputs via the supplied data.

This Exhibit shows that hallucination and deference to outdated information can be removed from even legacy Language Models via proper RAG pipeline and accurate referential treatment of the supplied data. Both script driver code and User prompts can be refined to eliminate ambiguity and produce the desired analysis, leading to more-robust LLM utility. (QUERY #2) By grounding the LLM in the provided data and providing a schema by which to accurately Output information, RAG integration internally leads directly to factual reliability.

This demonstration replicates enterprise-grade RAG workflows: retrieval, schema enforcement, and factual grounding, all reproducible via the included code and data. This, as aforementioned, can, of course, be scaled to any desired level as required by a given User and / or their organization(s).